# Jason Yu

925-768-0036 jjjyyy101@outlook.com www.linkedin.com/in/JasonYu101 jasonyu0705

## Education

**California Polytechnic State University, San Luis Obispo** Expected May 2027

*Bachelor of Science in Computer Engineering (Major GPA: 3.89) (Presidents List)* *San Luis Obispo, CA*

- **Relevant Coursework:** Computer Architecture and Design, Data Structures, Circuits, Microcontrollers, Systems Programming, AI, GPU Architecture, HRI, Signals and Systems, Networks, Machine Learning in Embedded Systems

## Experience

**Process Control & Automation Engineer Intern** Sept. 2026 – Present

*Marathon Petroleum Corporation* | *Allen Bradley PLC, Ladder Logic, HMI, SCADA, Shell Scripting* *San Antonio, TX*

- Designing and testing **PLC/SCADA** control systems using ladder logic programming in **Studio5000**, HMI screens in **Crimson 3.1 & 3.2**, and I/O testing to support industrial automation and plant equipment upgrades
- Developing a **GraphicAccess API** tool to generate and modify **AVEVA System Platform SCADA** screens via XML
- Writing shell scripts to poll data from DA50 protocol converter and automate PLC file searching & classification

**Embedded Software/Firmware Engineer Intern** Jun. 2026 – Sept. 2026

*Molex* | *SPI, Embedded C, GitLab, Test case development, Python, Verilog, Cadence Allegro, JTAG* *Fremont, CA*

- Architected fault-insertion firmware for optical gain block, allowing alarm coverage **validation across 5 faults**
- Redesigned snapshot format for new optical gain block hardware; validated via **SPI** and **UART** with **50+ tests**
- Built and integrated an embedded test bed used by 10+ engineers to validate optical EDFA hardware
- Replaced legacy **logarithmic amplifier IC** in circuit, allowing for the addition of **dark current compensation**

**Software Engineer Intern** Sept. 2025 – Dec. 2025

*Marathon Petroleum Corporation* | *C#, .NET, Azure Functions, Blazor, Git, Azure Fileshare, Scrum* *San Antonio, TX*

- Migrated legacy app from IaaS to PaaS and automated financial workflows using **C#**, **Azure Functions**, and **ML.NET**.

**Software Engineer Intern** Jun. 2025 – Aug. 2025

*Hunter Industries* | *AngularJS, React, AWS Lambda & RDS, JavaScript, Typescript, CSS, Git, SQL, Agile* *San Marcos, CA*

- Developed data-export feature for irrigation platform using **AWS Lambda, Amazon RDS**, and **AngularJS**

## Projects

**UDP File Transfer Emulator** | *Networking, UDP, Client-Server Communication* Mar. 2026

- Implemented reliable UDP file transfer in C with a sliding window and selective reject under simulated packet loss

**Magnetic Stripe Emulator** | *Microcontrollers, I2C, EEPROM, C, H-Bridge, Solenoid, UART* Jun. 2025

- Developed an **STM32-based** device capable of emulating magnetic card swipes using a solenoid and motor driver, implementing robust **C** code for a seamless user experience
- Researched and designed a circuit to replicate magnetic stripe specifications, allowing for successful data transmission to standard card readers via induced magnetic fields

**Nine Volt Battery Charger** | *Microcontrollers, Circuit Design, UART, ADC* Jun. 2025

- Built a 9V charger using a step-down **transformer**, **full-wave bridge rectifier**, and **BJT-based current regulator** to achieve constant-current/constant-voltage (CC/CV) tapering
- Integrated an **STM32** to monitor battery voltage via a voltage divider and low-pass filter, using the **ADC** to estimate state-of-charge in real time and report it over **UART**

**FSK Receiver-Transmitter System** | *Circuit Theory, Circuit Design, Analog Electronics* Mar. 2025

- Built a 555 timer-based IR transmitter and photodiode-based receiver with a **transimpedance amplifier** and analog tachometer circuit (comparator, one-shot, low-pass filter) to decode frequencies into distinct DC voltage levels

**Snake in RISC-V Assembly** | *Verilog, FPGA programming, VGA Manipulation, Assembly Coding* Dec. 2024

- Utilized the previously constructed processor with an added VGA module and RISC-V assembly code as the base to create a rendition of snake, with I/O control inputs as buttons on a BASYS 3 board

## Technical Skills

**Software Skills**: Java, Python, System Verilog, Assembly, C, C++, LTSpice, Vivado, JavaScript, Git, Jest, Jasmine
**Hardware Skills**: Oscilloscope, Multimeter, Soldering, Arduino & STM32, I²C/SPI/UART, MATLAB, Wireshark
**Languages**: English, Chinese, French